Convergence in probability and Relative entropy method for moderate interacting system

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Setting and literature

Stochastic moderate interacting system

Model with N-particle moderate interacting system

$$dX_i^{N,\varepsilon} = \frac{1}{N}\sum_{j=1}^{N} \nabla\tilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon})\,dt + \sqrt{2}dW_t^i$$
$$X_i^N(0) = \xi_i \in \mathbb{R}^d,\ \text{i.i.d},\ \rho_0 \text{ is the probability density function of } \xi_i.$$

where $\tilde{\Phi}_\varepsilon$ is smoothed version of given singular interaction potential Φ.

Example 2-D Keller-Segel system. $\Phi(x) = -\frac{1}{2\pi}\ln|x|$, $\tilde{\Phi}_\varepsilon$ can be given by

$$\widetilde{\Phi} = \begin{cases} -\frac{1}{2\pi}\ln|x|, & |x| > 2\varepsilon, \\ -\frac{1}{4\pi\varepsilon}|x| + \frac{1}{2\pi}(1-\ln 2\varepsilon), & |x| \le 2\varepsilon, \end{cases} \qquad \widetilde{\Phi}_\varepsilon = j_\varepsilon * \widetilde{\Phi}.$$

The corresponding ε-McKean-Vlasov equation (fix ε and $N \to \infty$) is

$$d\bar{X}_i^\varepsilon = \nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon)\,dt + \sqrt{2}\,dW_t^i.$$

where the law of $\bar{X}_i^\varepsilon$ satisfies

$$\partial_t\rho^\varepsilon - \Delta\rho^\varepsilon = \nabla\cdot(\rho^\varepsilon\nabla c^\varepsilon), \qquad c^\varepsilon = \widetilde{\Phi}_\varepsilon * \rho^\varepsilon,$$

Expected mean-field limit with scalling $\varepsilon(N) \to 0$ as $N \to \infty$

McKean-Vlasov equation

$$d\hat{X}_i = \nabla\Phi * \rho(\hat{X}_i)\, dt + \sqrt{2}\, dW_t^i.$$

where the law of $\hat{X}_i$ satisfies

$$\partial_t \rho - \Delta\rho = \nabla \cdot (\rho \nabla c), \qquad c = \Phi * \rho,$$

Benefit from PDE solutions:
Compare to the classical work in mean-field limit (measure valued solutions).
$\nabla\Phi * \rho$ can be bounded Lipschitz for "good" ρ.

Remark: It is known that log scaling $\varepsilon \sim 1/\log N$ is "almost" trivial (proof similar to the bounded Lipschitz case). Reason: "too much" cut-off.

Goal: Algebraic scaling $\varepsilon = N^{-\beta}$ for some $\beta > 0$.

Selected Literature Review

Mean field limit (for non-Lipschitz potentials)

- K. Oelschläger (1987): repulsive δ, L^2 estimate with rate $o(N^{-1/2})$
- Lazarovici and Pickl (2017): Vlasov, Convergence in probability
- Jabin and Wang, (2018): relative entropy
- S. Serfaty (2020): Coulomb potential, modulated energy
- Rosenzweig and Serfaty (2023): Reisz potential
- Olivera, Richard, and Tomasevic (2023): Semigroup approach
- ......

Main result and proof ideas

Consider the high dimensional PDEs with factorized initial data $\rho_0^{\otimes N}$.

$$\partial_t \rho^{N,\varepsilon} - \sum_{i=1}^N \Delta_{x_i} \rho^{N,\varepsilon} + \sum_{i=1}^N \nabla_{x_i} \Big(\rho^{N,\varepsilon} \frac{1}{N} \sum_{j=1}^N \nabla \widetilde{\Phi}_\varepsilon (x_i - x_j)) \Big) = 0$$

$$\partial_t \rho^{\otimes N} - \sum_{i=1}^N \Delta_{x_i} \rho^{\otimes N} + \sum_{i=1}^N \nabla_{x_i} \Big(\rho^{\otimes N} \nabla \Phi * \rho(x_i) \Big) = 0$$

where $\rho^{N,\varepsilon}$ is the joint law of $X_i^{N,\varepsilon}, 1 \leq i \leq N$.

k- marginal of $\rho^{N,\varepsilon}$, $1 \leq k \leq N$ is given by

$$\rho^{N,\varepsilon;k}(x_{k+1}, \cdots, x_N) = \int_{\mathbb{R}^{d(N-k)}} \rho^{N,\varepsilon}(x_1, \cdots, x_N) dx_{k+1} \cdots dx_N.$$

Theorem (Mean field limit)

Let $\rho^{N,\varepsilon;1}$ be the 1-particle marginal of $\rho^{N,\varepsilon}$ and ρ be the solution of the limiting PDE, then $\forall \beta \in (0, \beta_1)$ ($\beta_1 = \frac{1}{8}$ for K-S), it holds

$$\|\rho^{N,\varepsilon;1} - \rho\|_{L^1(\mathbb{R}^d)} \leq \frac{C(t)}{N^\beta}.$$

Idea:

Convergence in probability	"+"	Relative entropy
(Originally: Lazarovici and Pickl)		(Originally: Jabin and Wang)

New: Mean field limit in L^1 norm for moderately interacting system with Coulomb potential on $\mathbb{R}^d$.

Key Step: Use convergence in probability to close the relative entropy estimate for moderate interacting systems.

Convergence in probability

Proposition (Convergence in probability)

For any $\alpha \in (0, \frac{1}{2})$ *satisfies* $0 < \beta < \min\{\frac{\alpha}{3}, \frac{1}{4}, \frac{1}{2} - \alpha\}$, *then* $\forall \gamma > 0$ *it holds*

$$\sup_{0 \leq t \leq T} \mathbb{P}\left(\max_{1 \leq i \leq N} \left| X_i^{N,\varepsilon} - \bar{X}_i^{\varepsilon} \right| \geq N^{-\alpha} \right) \leq \frac{C(\gamma)}{N^{\gamma}}.$$

Idea: Stopping time, law of large numbers (Lazarovici and Pickl)

A key tool: A version of law of large numbers

Lemma

let $(\bar{Y}_i)_{i=1,\cdots,N}$ *be i.i.d random variables,* $v \in L^1(\mathbb{R}^2)$ *be their density,* $U \in L^\infty(\mathbb{R}^2)$ *be given, Then for any* $\widetilde{k} \geq 1$ *, we have:*

$$\mathbb{P}(\mathcal{A}_\theta^N(U,v)) \leq N \max_{1\leq i\leq N} \mathbb{P}(\mathcal{A}_\theta^i(U,v)) \leq N^{2\widetilde{k}(\theta-\frac{1}{2})+1} C(\widetilde{k}) \|U\|_{L^\infty}^{2\widetilde{k}}.$$

where $\mathcal{A}_\theta^N(U,v) = \cap_{i=1}^N \mathcal{A}_\theta^i(U,v)$ *with*

$$\mathcal{A}_\theta^i(U,v) = \left\{\omega \in \Omega : \left|\frac{1}{N}\sum_{j=1}^N U(\bar{Y}^i - \bar{Y}^j) - U * v(Y^i)\right| > \frac{1}{N^\theta}\right\}.$$

Estimate for the cut-offed potential

Lemma (2-D K-S)

$$\||\nabla^2\widetilde{\Phi}| * g\|_{L^\infty} \leq C(\|g\|_{L^1} - \ln 2\varepsilon \|g\|_{L^\infty}).$$

Stopping time

Let $X_i^{N,\varepsilon}$ and $\bar{X}_i^{\varepsilon}$ be the solution of particle system and the ε- Mckean Vlasov equation. We define the stopping time $\tau(\omega)$ by

$$\tau(\omega) = \inf\{t \in (0, T] | \max_{1\le i\le N} |X_i^{N,\varepsilon}(t) - \bar{X}_i^{\varepsilon}(t)| \ge N^{-\alpha}\}.$$

and introduce the cut-off process $S(\omega, t)$

$$S(\omega, t) = N^{2\alpha k} \max_{1\le i\le N} |(X_i^{N,\varepsilon} - \bar{X}_i^{\varepsilon})(t \wedge \tau)|^{2k} \le 1.$$

where $k \in \mathbb{N}$ is to be determined later. Markov's inequality implies that

$$\sup_{0\le t\le T} \mathbb{P}(\max_{1\le i\le N} |X_i^{N,\varepsilon} - \bar{X}_i^{\varepsilon}| \ge N^{-\alpha}) \le \sup_{0\le t\le T} \mathbb{P}(S_t = 1) \le \sup_{0\le t\le T} \mathbb{E}(S_t).$$

Evolution of $S(t)$

Take the difference, apply Ito's formula, and take the expectation,

$$\mathbb{E}(S(t)) = N^{2\alpha k}\mathbb{E}(\max_{1\leq i\leq N} |(X_i^{N,\varepsilon} - \bar{X}_i^{\varepsilon})(t\wedge\tau(\omega))|^{2k})$$

$$\leq CN^{\alpha}\mathbb{E}\Big(\max_{1\leq i\leq N}\int_0^{t\wedge\tau} |\nabla\widetilde{\Phi}_{\varepsilon} * \mu_N(X_i^{N,\varepsilon}(s)) - \nabla\widetilde{\Phi}_{\varepsilon} * \bar{\mu}_N(\bar{X}_i^{\varepsilon}(s))|S(s)^{\frac{2k-1}{2k}}\, ds\Big)$$

$$+ CN^{\alpha}\mathbb{E}\Big(\max_{1\leq i\leq N}\int_0^{t\wedge\tau} |\nabla\widetilde{\Phi}_{\varepsilon} * \bar{\mu}_N(\bar{X}_i^{\varepsilon}(s)) - \nabla\widetilde{\Phi}_{\varepsilon} * \rho^{\varepsilon}(\bar{X}_i^{\varepsilon}(s))|S(s)^{\frac{2k-1}{2k}}\, ds\Big)$$

$$:= K_1 + K_2,$$

where

$$\mu_N = \frac{1}{N}\sum_{j=1}^{N}\delta_{X_i^{N,\varepsilon}(s)}, \qquad \bar{\mu}_N = \frac{1}{N}\sum_{j=1}^{N}\delta_{\bar{X}_i^{\varepsilon}(s)}$$

are the empirical measures.

Estimate for K_2, law of large numbers Let

$$A_{\theta_1}(\nabla\widetilde{\Phi}_\varepsilon, \rho^\varepsilon) = \cup_{i=1}^N A_{\theta_1}^i,$$

and obtain the estimate for K_2

$$K_2 \le CN^\alpha \mathbb{E}\Big(\max_{1\le i\le N} \int_0^{t\wedge\tau} |\nabla\widetilde{\Phi}_\varepsilon * \bar{\mu}_N(\bar{X}_i^\varepsilon(s)) - \nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon(s))| S(s)^{\frac{2k-1}{2k}}\, ds\Big)$$

$$\le CN^{2\alpha k}\mathbb{E}\left(\max_i \int_0^{t\wedge\tau} (\mathbb{I}_{A_{\theta_1}^c} + \mathbb{I}_{A_{\theta_1}}) \Big(\frac{1}{N}\sum_{j=1}^N \nabla\widetilde{\Phi}_\varepsilon * \bar{\mu}_N - \nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon\Big)^{2k} (\bar{X}_i^\varepsilon) ds\right)$$

$$+ C\int_0^t \mathbb{E}(S(s))\, ds$$

$$\le CN^{2\alpha k}\Big(\frac{1}{N^{2\theta_1 k}} + \|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty}^{2k}\mathbb{P}(A_{\theta_1})\Big) + C\int_0^t \mathbb{E}(S(s))\, ds$$

$$\le C\big(N^{(\alpha-\theta_1)2k} + N^{2\alpha k}C(\widetilde{k})\|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty}^{2(k+\widetilde{k})} N^{2\widetilde{k}(\theta_1-\frac{1}{2})+1}\big) + \int_0^t \mathbb{E}(S(s))\, ds.$$

Estimate for K_1

By using Taylor's expansion at point $\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon$ we have

$$K_1 \leq CN^\alpha \mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \Big| \frac{1}{N} \sum_{j=1}^N \nabla \widetilde{\Phi}_\varepsilon (X_i^{N,\varepsilon} - X_j^{N,\varepsilon}) - \nabla \widetilde{\Phi}_\varepsilon (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon) \Big| S^{\frac{2k-1}{2k}}\, ds \Big)$$

$$\leq CN^\alpha \mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \Big| \frac{1}{N} \sum_{j=1}^N \nabla^2 \widetilde{\Phi}_\varepsilon (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)(X_i^{N,\varepsilon} - X_j^{N,\varepsilon} - (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)) \Big| S^{\frac{2k-1}{2k}}\, ds \Big)$$

$$+ CN^\alpha \mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \|\nabla^3 \widetilde{\Phi}_\varepsilon\|_{L^\infty} \Big| \frac{1}{N} \sum_{j=1}^N (X_i^{N,\varepsilon} - X_j^{N,\varepsilon} - (\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)) \Big|^2 S^{\frac{2k-1}{2k}}\, ds \Big)$$

$$:= K_{11} + K_{12}.$$

where K_{12} can be estimated by

$$K_{12} \leq \frac{C}{N^\alpha} \|\nabla^3 \widetilde{\Phi}_\varepsilon\|_{L^\infty} \int_0^t \mathbb{E}(S(s))\, ds.$$

Estimate for K_{11}

For K_{11}, we need to use the law of large numbers

$$\begin{aligned}
K_{11} \leq & CN^{\alpha}\mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \frac{1}{N}\sum_{j=1}^{N} |\nabla^2\widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| \max_i |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| S^{\frac{2k-1}{2k}}\, ds\Big) \\
\leq & C\mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \frac{1}{N}\sum_{j=1}^{N} (|\nabla^2\widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2\widetilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon))Sds\Big) \\
& + C\mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} (|\nabla^2\widetilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon))Sds\Big) \\
=: & K_{111} + K_{112},
\end{aligned}$$

where

$$K_{112} \leq C\||\nabla^2\widetilde{\Phi}_\varepsilon| * \rho^\varepsilon\|_{L^\infty} \int_0^t \mathbb{E}(S(s))\, ds \leq C(1 - \ln 2\varepsilon) \int_0^t \mathbb{E}(S(s))\, ds.$$

Estimate for K_{111}, law of large numbers

Take $U = |\nabla^2 \widetilde{\Phi}_\varepsilon|, \ v = \rho^\varepsilon$ and

$$A_\theta(|\nabla^2 \widetilde{\Phi}_\varepsilon|, \rho^\varepsilon) = \cup_{i=1}^N A_\theta^i \Rightarrow A_\theta^c(|\nabla^2 \widetilde{\Phi}_\varepsilon|, \rho^\varepsilon) = \cap_{i=1}^N (A_\theta^i)^c.$$

then with $\theta = 0$ we have

$$K_{111} \leq C\mathbb{E}\Big(\max_i \int_0^{t\wedge\tau} \frac{1}{N} \sum_{j=1}^N (|\nabla^2 \widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2 \widetilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon)) S ds \Big)$$

$$\leq C\mathbb{E}\Bigg((\mathbb{I}_{A_\theta^c} + \mathbb{I}_{A_\theta}) \max_i \int_0^{t\wedge\tau} \Big(\frac{1}{N} \sum_{j=1}^N |\nabla^2 \widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)| - |\nabla^2 \widetilde{\Phi}_\varepsilon| * \rho^\varepsilon(\bar{X}_i^\varepsilon) \Big) S ds \Bigg)$$

$$\leq \frac{C}{N^\theta} \int_0^t \mathbb{E}(S(s))\, ds + C \|\nabla^2 \widetilde{\Phi}_\varepsilon\|_{L^\infty} \mathbb{P}(A_\theta)$$

$$\leq C \int_0^t \mathbb{E}(S(s))\, ds + C \|\nabla^2 \widetilde{\Phi}_\varepsilon\|_{L^\infty} \|\nabla^2 \widetilde{\Phi}_\varepsilon\|_{L^\infty}^{2\widetilde{k}} N^{2\widetilde{k}(\theta - \frac{1}{2})+1}.$$

Gronwall's inequality

Notice for 2-D K-S, $\|\nabla^3\widetilde{\Phi}_\varepsilon\|_{L^\infty} \le \varepsilon^{-3}$, $\|\nabla^2\widetilde{\Phi}_\varepsilon\|_{L^\infty} \le \varepsilon^{-2}$, $\|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty} \le \varepsilon^{-1}$. we have

$$\begin{aligned}
\mathbb{E}(S(t)) \le & C\int_0^t \Big(1 + N^{-\alpha}\|\nabla^3\widetilde{\Phi}_\varepsilon\|_{L^\infty} - \ln 2\varepsilon\Big)\mathbb{E}(S(s))ds + CN^{(\alpha-\theta_1)2k} \\
& + CN^{2\widetilde{k}(\theta-\frac{1}{2})+1}\|\nabla^2\widetilde{\Phi}_\varepsilon\|_{L^\infty}^{2\widetilde{k}+1} + CN^{2\alpha k+2\widetilde{k}(\theta_1-\frac{1}{2})+1}\|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty}^{2(k+\widetilde{k})} \\
\le & C\int_0^t \Big(1 + N^{3\beta-\alpha} + \ln N^\beta\Big)\mathbb{E}(S(s))\,ds + CN^{(\alpha-\theta_1)2k} \\
& + CN^{2\widetilde{k}(-\frac{1}{2}+2\beta)+(2\beta+1)} + CN^{2k(\alpha+\beta)+2\widetilde{k}(\theta_1-\frac{1}{2}+\beta)+1}.
\end{aligned}$$

Therefore for $0 < \alpha < \frac{1}{2}$, we take $0 < \beta < \min\{\frac{\alpha}{3}, \frac{1}{4}, \frac{1}{2} - \alpha\}$ and $\theta_1 \in (\alpha, \frac{1}{2} - \beta)$. Then $\forall\gamma > 0$, there exists k and $\widetilde{k}$ s.t.

$$\mathbb{E}(S(t)) \le CN^{-2\gamma} + C\int_0^t \Big(1 + N^{3\beta-\alpha} + \ln N^\beta\Big)\Big)\mathbb{E}(S(s))\,ds.$$

Relative entropy

A key tool: Relative entropy method Consider the high dimensional PDEs with factorized initial data $\rho_0^{\otimes N}$.

$$\partial_t \rho^{N,\varepsilon} - \sum_{i=1}^N \Delta_{x_i} \rho^{N,\varepsilon} + \sum_{i=1}^N \nabla_{x_i} \Big(\rho^{N,\varepsilon} \big(\frac{1}{N} \sum_{j=1}^N \nabla \widetilde{\Phi}_\varepsilon (x_i - x_j) - \nabla \widetilde{\Phi}_\varepsilon * \widetilde{f}_N(x_i) \big) \Big) = 0$$

$$\partial_t \rho^{\otimes N} - \sum_{i=1}^N \Delta_{x_i} \rho^{\otimes N} + \sum_{i=1}^N \nabla_{x_i} \Big(\rho^{\otimes N} \big(\nabla \Phi * \rho(x_i) - \nabla \Phi * \widetilde{f}_N f(x_i) \big) \Big) = 0$$

The relative entropy of N-particle system $\mathcal{H}$ defined by

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) = \frac{1}{N} \mathcal{H}_N(\rho^{N,\varepsilon} | \rho^{\otimes N}) = \frac{1}{N} \int_{\mathbb{R}^{2N}} \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \cdot \rho^{\otimes N} \, dX_N .$$

Proposition (Relative entropy estimate (Using convergence in probability))

If $\varepsilon = N^{-\beta}$, $0 < \beta < \beta_1 (= \frac{1}{8})$ for $2-D$ KS, then

$$\mathcal{H}(\rho^{N,\varepsilon} | \rho^{\otimes N}) \le \frac{C(t)}{N^{2\beta}},$$

Compute the time evolution of the relative entropy, we have

$$\begin{aligned}
&\frac{d}{dt}\mathcal{H}(\rho^{N,\varepsilon}|\rho^{\otimes N}) + \frac{1}{N}\int_{\mathbb{R}^{2N}} \sum_{i=1}^{N} \left|\nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}}\right|^2 \rho^{N,\varepsilon} dX_N \\
=&\int_{\mathbb{R}^{2N}} \frac{1}{N}\sum_{i=1}^{N}\Big(\nabla\Phi * \rho(x_i) - \frac{1}{N}\sum_{j=1}^{N}\nabla\widetilde{\Phi}_\varepsilon(x_i - x_j)\Big)\nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}} \rho^{N,\varepsilon}\, dX_N \\
\leq&\frac{1}{2N}\int_{\mathbb{R}^{2N}} \sum_{i=1}^{N} \left|\nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}}\right|^2 \rho^{N,\varepsilon} dX_N \\
&+ \frac{1}{2}\int_{\mathbb{R}^{2N}} \frac{1}{N}\sum_{i=1}^{N}\left|\nabla\Phi * \rho(x_i) - \frac{1}{N}\sum_{j=1}^{N}\nabla\widetilde{\Phi}_\varepsilon(x_i - x_j)\right|^2 \rho^{N,\varepsilon}\, dX_N.
\end{aligned}$$

As a summary,

$$\begin{aligned}
&\frac{d}{dt}\mathcal{H}(\rho^{N,\varepsilon}\big|\rho^{\otimes N}) + \frac{1}{2N}\int_{\mathbb{R}^{2N}} \sum_{i=1}^{N} |\nabla_{x_i} \log \frac{\rho^{N,\varepsilon}}{\rho^{\otimes N}}\Big|^2 \rho^{N,\varepsilon}\, dX_N \\
\leq&\frac{1}{2}\mathbb{E}\Big(\frac{1}{N}\sum_{i=1}^{N}\Big|\nabla\Phi * \rho(X_i^{N,\varepsilon}) - \frac{1}{N}\sum_{j=1}^{N}\nabla\widetilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon})\Big|^2\Big) := M.
\end{aligned}$$

Use convergence in probability to close the estimate of M

$$
\begin{aligned}
2M \le & \mathbb{E}\Big(\frac{1}{N}\sum_i |\nabla\Phi * \rho(X_i^{N,\varepsilon}) - \nabla\Phi_\varepsilon * \rho(X_i^{N,\varepsilon})|^2\Big) \quad \le \varepsilon^2(1+\|\rho\|^2_{W^{1,q}}) \\
&+ \mathbb{E}\Big(\frac{1}{N}\sum_i |\nabla\Phi_\varepsilon * \rho(X_i^{N,\varepsilon}) - \nabla\widetilde{\Phi}_\varepsilon * \rho(X_i^{N,\varepsilon})|^2\Big) \quad \le \varepsilon^2(1+\|\rho\|^2_{W^{1,q}}) \\
&+ \mathbb{E}\Big(\frac{1}{N}\sum_i |\nabla\widetilde{\Phi}_\varepsilon * \rho(X_i^{N,\varepsilon}) - \nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(X_i^{N,\varepsilon})|^2\Big) \quad \le \varepsilon^2(1+\|\rho\|^2_{W^{1,q}}) \\
&+ \mathbb{E}\Big(\frac{1}{N}\sum_i |\nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(X_i^{N,\varepsilon}) - \nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon)|^2\Big) \\
&+ \mathbb{E}\Big(\frac{1}{N}\sum_i |\nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon(\bar{X}_i^\varepsilon) - \frac{1}{N}\sum_{j=1}^N \nabla\widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon)|^2\Big) \\
&+ \mathbb{E}\Big(\frac{1}{N}\sum_i |\frac{1}{N}\sum_{j=1}^N (\nabla\widetilde{\Phi}_\varepsilon(\bar{X}_i^\varepsilon - \bar{X}_j^\varepsilon) - \nabla\widetilde{\Phi}_\varepsilon(X_i^{N,\varepsilon} - X_j^{N,\varepsilon})|^2)\Big) \\
:= & \sum_{l=1}^6 M_l.
\end{aligned}
$$

Since $\bar{X}_i^\varepsilon$ are i.i.d. random variables, we know

$$M_5 \leq \frac{1}{N}\|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty}^2 \leq N^{2\beta-1}. \quad \text{law of large numbers}$$

By the result of convergence in probability, we have for set

$$\mathcal{A}_\alpha = \{\omega \in \Omega | \max_{1\leq i\leq N} |X_i^{N,\varepsilon} - \bar{X}_i^\varepsilon| > N^{-\alpha}\},$$

the estimate

$$\mathbb{P}(\mathcal{A}_\alpha) \leq \frac{C(\gamma)}{N^\gamma}.$$

Therefore, $M_4 + M_6$

$$\begin{aligned}
&\leq C(\gamma)\frac{\|\nabla\widetilde{\Phi}_\varepsilon\|_{L^\infty}^2 + \|\nabla\widetilde{\Phi}_\varepsilon * \rho^\varepsilon\|_{L^\infty}^2}{N^\gamma} + \frac{\|D^2\widetilde{\Phi}_\varepsilon\|_{L^\infty}^2 + \|D^2\widetilde{\Phi}_\varepsilon * \rho^\varepsilon\|_{L^\infty}^2}{N^{2\alpha}} \\
&\leq C(\gamma)\frac{\varepsilon^2}{N^\gamma} + \frac{\varepsilon^4}{N^{2\alpha}} \leq C(N^{2\beta-\gamma} + N^{4\beta-2\alpha}) \leq CN^{-\frac{2\alpha}{3}}.
\end{aligned}$$

The L^1 convergence follows from:

- Relative entropy estimate:

$$\mathcal{H}(\rho^{N,\varepsilon}|\rho^{\otimes N}) \leq C \min\{N^{-\beta}, N^{-\frac{2\alpha}{3}}\} \to 0.$$

- Super-Additivity of Relative Entropy:

$$\mathcal{H}(\rho^{N,\varepsilon}|\rho^{\otimes N}) = \frac{1}{N}\mathcal{H}_N(\rho^{N,\varepsilon}|\rho^{\otimes N}) \geq \frac{1}{2k}\mathcal{H}_k(\rho^{N,\varepsilon;k}|\rho^{\otimes k})$$

- Csiszàr-Kullback-Pinsker:

$$\|f - g\|^2_{L^1(\mathbb{R}^d)} \leq 2\mathcal{H}(f|g)$$

Therefore, we obtain

$$\|\rho^{N,\varepsilon;1} - \rho\|_{L^1(\mathbb{R}^d)} \to 0.$$

Possible generalizations

- Second order system (Vlasov Poisson, Cucker-Smale)
- Convergence in probability for weaker norms
- Fermionic quantum systems
- Further fluctuation estimates
- Derivation of cross-diffusion systems
- ......

Happy Birthday, Shih-Hsien !